One Edit

AlgoExpert · Medium · Strings

The problem

- Are two strings at most one insert/delete/replace apart?

Intuition

- Walk both strings; at the first mismatch, try skipping in the longer one or replacing, once.

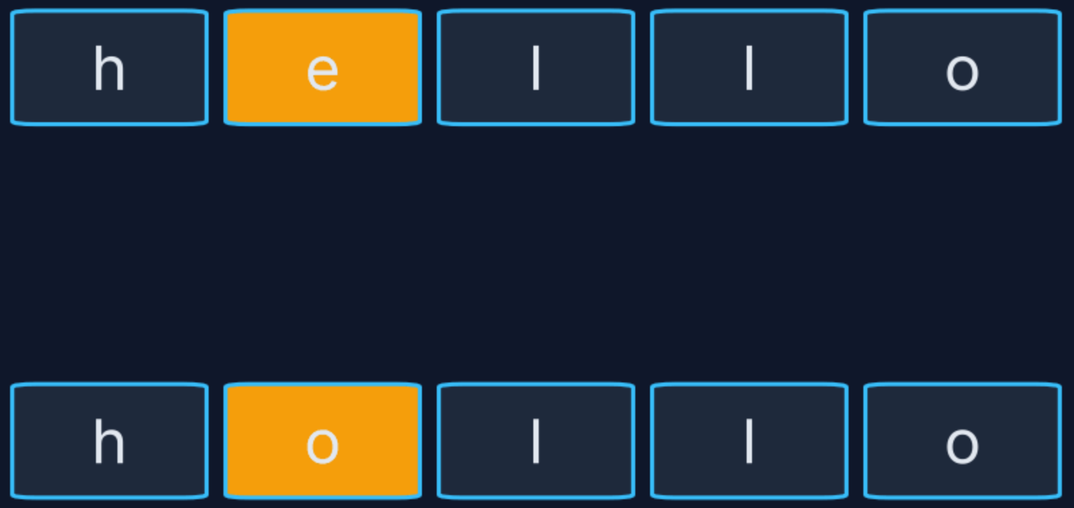
Brute force vs optimal

- Compute full edit distance → O(n·m)
- Single scan with one allowed mismatch → O(n)

Approach

- 1. If lengths differ by more than one, false
- 2. Find the first mismatch
- 3. Compare the remaining suffixes

Worked example




```
one replace → true
```

'hello' vs 'hollo': one replace → true

Worked example






```
one delete → true
```

'hello' vs 'helo': one delete → true

Worked example






```
two mismatches → false
```

'hello' vs 'hlelo': two mismatches → false

Java solution

```java
public class OneEdit {
    public static boolean oneEdit(String a, String b) {
        if (Math.abs(a.length() - b.length()) > 1) return false;
        int i = 0, j = 0;
        boolean edited = false;
        while (i < a.length() && j < b.length()) {
            if (a.charAt(i) != b.charAt(j)) {
                if (edited) return false;
                edited = true;
                if (a.length() > b.length()) i++;
                else if (a.length() < b.length()) j++;
                else { i++; j++; }
            } else { i++; j++; }
        }
        return true;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Edge cases & pitfalls

- Length differs by > 1 → false
- Identical strings → true (zero edits)
- Mismatch at the end

Interview tips & follow-ups

- Relate to Levenshtein distance

Recap

- Walk both strings; at the first mismatch, try skipping in the longer one or replacing, once.
- Complexity: O(n)